3D Interactive Map of Kolhapur

A MINI PROJECT-I REPORT

Submitted by

Anusha Patil	**(2425000174) (B-44)**
Pratik Patil	**(2425000705) (B-46)**
Adarsh Gurav	**(2425000776) (B-73)**
Sanket Magdum	**(2425000540) (B-77)**

in partial fulfillment for the award of the degree of

BACHELORS OF TECHNOLOGY

IN

COMPUTER SCIENCE ENGINEERING



KOLHAPUR INSTITUTE OF TECHNOLOGY'S

COLLEGE OF ENGINEERING KOLHAPUR

(EMPOWERED AUTONOMOUS)

OCTOBER 2025

KOLHAPUR INSTITUTE OF TECHNOLOGY'S
COLLEGE OF ENGINEERING KOLHAPUR
(EMPOWERED AUTONOMOUS)

CERTIFICATE

This is to certify that the Mini Project-I report entitled, **" 3D INTERACTIVE MAP OF KOLHAPUR "** submitted by **" Anusha Patil (44), Pratik Patil (46) , Adarsh Gurav (73), Sanket Magdum (77)"**, in partial fulfillment for the award of the degree of **" BACHELORS OF TECHNOLOGY "** in **" COMPUTER SCIENCE AND ENGINEERING "** at Kolhapur Institute of Technology's College of Engineering Kolhapur (Empowered Autonomous), Kolhapur, Maharashtra, INDIA, is a record of his / her own work carried out under my / our supervision and guidance.

SIGNATURE

Dr. Lingaraj Hadimani
HEAD OF THE DEPARTMENT

SIGNATURE

Mr. Vivek Ugale
SUPERVISOR

KOLHAPUR INSTITUTE OF TECHNOLOGY'S
COLLEGE OF ENGINEERING KOLHAPUR
(EMPOWERED AUTONOMOUS)

DECLARATION

I hereby declare that the Mini Project-I entitled, **" 3D INTERACTIVE MAP OF KOLHAPUR "** submitted to Kolhapur Institute of Technology's College of Engineering Kolhapur (Empowered Autonomous), Kolhapur, Maharashtra, INDIA in the partial fulfillment of the award of the Degree of **" BACHELORS OF TECHNOLOGY "** in **" COMPUTER SCIENCE AND ENGINEERING "** is a Bonafide work carried out by me. The material contained in this Project has not been submitted to any University or Institution for the award of any degree.

NAME OF THE STUDENT(S)

Anusha Patil. (B-44) (2425000174)
Pratik Patil. (B-46) (2425000705)
Adarsh Gurav. (B-73) (2425000776)
Sanket Magdum. (B-77) (2425000540)

Place: Kolhapur.
Date:

KOLHAPUR INSTITUTE OF TECHNOLOGY'S
COLLEGE OF ENGINEERING KOLHAPUR
(EMPOWERED AUTONOMOUS)

ACKNOWLEDGEMENT

We take this opportunity to express our sincere gratitude to all those who supported and guided us in the successful completion of our project titled **"INTERACTIVE 3D MAP SYSTEM FOR KOLHAPUR DISTRICT."** We are deeply thankful to our respected guide **Mr. Vivek Ugale** for his valuable guidance, suggestions, and constant encouragement throughout the development of this project. We also express our heartfelt thanks to our **Head of Department, Dr. Lingaraj Hadimani**, for providing the required facilities and a motivating environment for our work. Our sincere appreciation extends to all the **faculty members of the Department of Computer Science And Engineering** for their helpful feedback and assistance. Finally, we thank our **friends and family** for their continuous support and motivation, which greatly contributed to the successful completion of this project.

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Sanket Magdum. (B-77) (2425000540)

Place: Kolhapur.
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ABSTRACT

Our project presents a **3D Interactive Map of the Kolhapur District**, developed to provide users with an immersive and realistic virtual experience. The system features **real-time 3D visualization of landmarks** using **Mapbox** for mapping and **Blender** for creating detailed 3D models. Each location on the map includes **multi-language information** in **English, Hindi, and Marathi**, along with **audio support** to make the platform more accessible and user-friendly for a diverse audience. The interactive interface allows users to easily explore various locations and gain insightful information about the district's cultural and geographical aspects.

In addition, the project integrates **360° street views** from **Google Maps**, offering a lifelike experience of the surroundings and enhancing user engagement. By combining technology and tourism, this project aims to promote **virtual tourism** and support **educational purposes**, enabling users to explore the **rich cultural heritage and historical significance** of Kolhapur in an interactive 3D environment. It serves as a step forward in using modern web technologies to make regional tourism more informative, accessible, and visually appealing.

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Chapter 1

INTRODUCTION

The project titled **"Interactive 3D Map of Kolhapur District"** is a web-based application designed to provide users with an engaging and informative way to explore the Kolhapur district. This project integrates modern web technologies with advanced 3D visualization tools, enabling users to interact with a virtual three-dimensional map environment. It bridges the gap between traditional map viewing and immersive visualization by allowing real-time exploration of landmarks, cultural sites, and key locations across the district.

This interactive mapping platform is built using technologies such as **Mapbox API**, **React.js**, **HTML**, **CSS**, **JavaScript**, and **Blender**. Each technology contributes to different aspects of the system — React for user interaction, Mapbox for map rendering, and Blender for realistic 3D modeling. The overall system offers a rich and dynamic experience that educates users while also promoting digital tourism and accessibility for local and global audiences.

1.1 Project Category

The project falls under the category of **Internet-based web applications**, with a primary focus on visualization and user interaction. It utilizes the **Mapbox API** to embed maps and geographic data in a customized way, while React provides a responsive interface for smooth navigation. The system also incorporates web-based rendering for 3D objects, ensuring a fluid and immersive experience even within standard browsers.

Being an interactive and multimedia-oriented system, this project combines **front-end visualization and backend data management**. It belongs to the emerging domain of **Geo-visualization applications**, which merge spatial data with modern web design. The project's scope extends to tourism, education, and regional planning, thereby making it an effective digital solution for multiple sectors.

1.2 Objectives

The main objective of this project is to design a **user-friendly, interactive 3D map** that showcases Kolhapur district landmarks with realistic visualizations. The system aims to present detailed information in **multiple languages** — English, Hindi, Marathi, Kannada, and Telugu — to ensure inclusivity. The integration of audio narration further enhances accessibility, making the platform useful for visually impaired users and tourists unfamiliar with the local language.

Additionally, the project seeks to include a **360° virtual street view** to offer users a real-world experience directly through the website. Another key objective is to provide a lightweight, fast-loading interface that ensures smooth navigation between locations. These combined features create a comprehensive digital exploration platform that represents the cultural and historical richness of Kolhapur district.

1.3 Problem Formulation

Traditional 2D maps, while useful, are often **static, text-heavy, and lack interactivity**, making them less engaging for users seeking an immersive experience. They do not allow users to visualize landmarks in three dimensions or access detailed information in their preferred language. This limits their use, especially for tourists or students who want an in-depth exploration of a region.

Therefore, there exists a **clear gap** in existing mapping platforms that fail to combine **custom 3D models, localization, and accessibility**. The proposed system addresses this issue by integrating realistic 3D environments, multilingual content, and audio features to make exploration both educational and enjoyable.

1.4 Identification / Reorganization of Need

The need for an interactive and visually engaging mapping system arises from the growing demand for **digital tourism, educational visualization, and cultural preservation**. A 3D interactive map can showcase important temples, monuments, and heritage locations in a manner that static maps cannot. This approach helps tourists plan their visits more efficiently while learning about each landmark.

For students and researchers, the platform serves as a valuable educational resource. It allows them to study the **geography, architecture, and cultural history** of Kolhapur district through visual representation and real-time interaction. By providing this data-rich visualization, the project fulfills a crucial requirement for modern, accessible geographic learning tools.

1.5 Existing System

Existing systems like **Google Maps** or **Apple Maps** provide excellent global navigation and 2D/3D views, but they lack **customized local detailing**. These platforms are not tailored for specific districts like Kolhapur and do not include locally modeled 3D landmarks or multilingual support. Additionally, they do not offer integrated audio narration or culturally specific data visualization.

As a result, users often experience limitations in terms of **local heritage representation** and **personalized interactivity**. This project bridges that gap by focusing specifically on Kolhapur's landmarks and adding layers of cultural, linguistic, and experiential detail that generic systems overlook.

1.6 Proposed System

The proposed system introduces a **customized 3D visualization** of Kolhapur district, featuring realistic 3D landmark models created using Blender. Each location is integrated into the Mapbox environment, allowing smooth navigation and interaction. Users can click on specific landmarks to open information windows that include descriptions, images, and audio narration in multiple languages.

Furthermore, the system supports **theme switching (light/dark modes)** for enhanced usability in different lighting conditions. The 360° **Street View integration** adds another level of realism, giving users a near-physical experience of being at the site. Collectively, these elements make the map dynamic, visually appealing, and functionally rich.

1.7 Unique Features

The uniqueness of the system lies in its **blend of interactivity, localization, and immersive design**. Unlike conventional mapping tools, it combines multilingual information, 3D models, and real-time visuals within a single web application. The addition of audio narration ensures inclusivity for differently-abled users and non-native speakers.

Moreover, features like **dark/light mode switching**, **responsive design**, and **360° view integration** provide an intuitive user experience across all devices. These innovative inclusions not only enhance usability but also make the project a pioneering step toward localized digital mapping and tourism promotion.

Chapter 2

REQUIREMENT ANALYSIS & SYSTEM SPECIFICATION

2.1 Feasibility Study

The proposed system is **technically feasible** because it uses reliable, open-source, and community-supported tools like **React**, **Mapbox**, and **Blender**. These technologies provide well-documented APIs and frameworks, making development efficient and error-free. The integration between these tools ensures compatibility across browsers and devices, while lightweight libraries make the system accessible even on mid-range hardware. The modular architecture of React allows easy expansion, ensuring that additional features such as real-time data or navigation paths can be incorporated later. From an **economic standpoint**, the project is also feasible because most of the tools used are free or offer educational access. Since the entire system runs through a web browser, there is no need for expensive hardware or dedicated software installations. The project is **operationally feasible** as well, as it delivers an intuitive, user-friendly interface that requires no technical training. Therefore, it is both cost-effective and practical for long-term use in academic and tourism-related environments.

2.2 Software Requirement Specification

The **functional requirements** of the system include displaying the Kolhapur district map in 3D, allowing users to click on landmarks to retrieve detailed information, providing multilingual content with audio narration, and integrating Google Street View for realistic visualization. The **non-functional requirements** focus on smooth performance, cross-browser compatibility, and efficient data loading. The user interface should be responsive and adaptive, ensuring consistent experience on desktops, tablets, and smartphones.

Security requirements involve **protection of API keys** (especially for Mapbox and Google APIs), ensuring unauthorized users cannot exploit them. The system follows a **modular structure**, where each feature (map, language, audio, theme) is a separate

component, ensuring maintainability and scalability. The backend stores landmark information and related resources, structured to minimize redundancy and maximize data retrieval speed.

2.3 Validation

The validation phase ensures that the project meets all functional and performance requirements. Each module was tested individually to verify that it performs its intended function. For instance, when users click on a landmark, the system correctly fetches data from the database and displays relevant information with accurate language translation and synchronized audio playback. Similarly, theme switching and 3D model rendering were validated to ensure no overlap or lag occurred during transitions.

Comprehensive testing was conducted using multiple browsers (Chrome, Edge, and Firefox) and devices (desktop and mobile) to confirm responsiveness and usability. The multilingual and audio systems were also verified through real-time user feedback. Validation confirmed that the platform operates smoothly and provides consistent output, ensuring an error-free, user-centric experience.

2.4 Expected Hurdles

Some challenges expected during development included integrating 3D models without performance degradation. Since web browsers have rendering limitations, optimization was necessary to ensure the map remains responsive even when multiple 3D objects are loaded. Another anticipated difficulty was managing **multilingual content** efficiently. Each language required separate data storage and synchronization with audio narration to maintain accuracy and timing.

Compatibility across browsers and devices was another hurdle. Different browsers handle WebGL and JavaScript differently, which could affect 3D performance. Efficient asset compression and asynchronous data loading were implemented to overcome these issues. Proper handling of large datasets, especially when embedding Street View and external APIs, was crucial for maintaining system stability.

2.5 SDLC Model

The project follows a **hybrid Software Development Life Cycle (SDLC)** model combining both **Waterfall** and **Iterative** approaches. The Waterfall approach provided a clear, linear process for major stages such as requirement analysis, design, and implementation, ensuring disciplined progress. Each stage was documented before moving to the next, reducing ambiguity and maintaining structured workflow.

Meanwhile, the iterative component allowed flexibility to refine certain modules based on feedback and testing outcomes. For instance, after initial testing, improvements were made in map interactions and multilingual data handling. This blend of structure and adaptability helped balance reliability with creativity, resulting in a robust and user-friendly system.

Chapter 3

SYSTEM DESIGN

3.1 Design Approach

The design follows a **function-oriented and modular architecture**, where each component has a specific purpose and operates independently. The modular design enhances reusability and simplifies maintenance, allowing future developers to easily add or modify features without affecting the entire system. Components like the Map Module, Audio System, and Theme Controller work together through well-defined interfaces.

This approach ensures a **clean separation of concerns**, making debugging and scalability easier. The frontend manages all user interactions, while APIs handle map data and rendering. The overall system design aims for a balance between performance, interactivity, and visual appeal, ensuring a seamless user experience even on resource-limited devices.

3.2 Detailed Design

The project comprises three main pages: the **Home Page**, **Map Page**, and **Information Page**. The Home Page introduces users to the system with an animated background and navigation menu that directs them to other sections. The Map Page contains the 3D interactive map, where users can click on various landmarks. The Information Page dynamically loads details, images, and audio narration for the selected location.

Each page is interconnected to ensure smooth navigation. The design also includes a **responsive layout** that adjusts to different screen sizes. The system employs reusable React components to maintain consistency in design. All pages are visually appealing, well-structured, and easy to navigate, focusing on accessibility and engagement.

3.3 Structured Design Tools

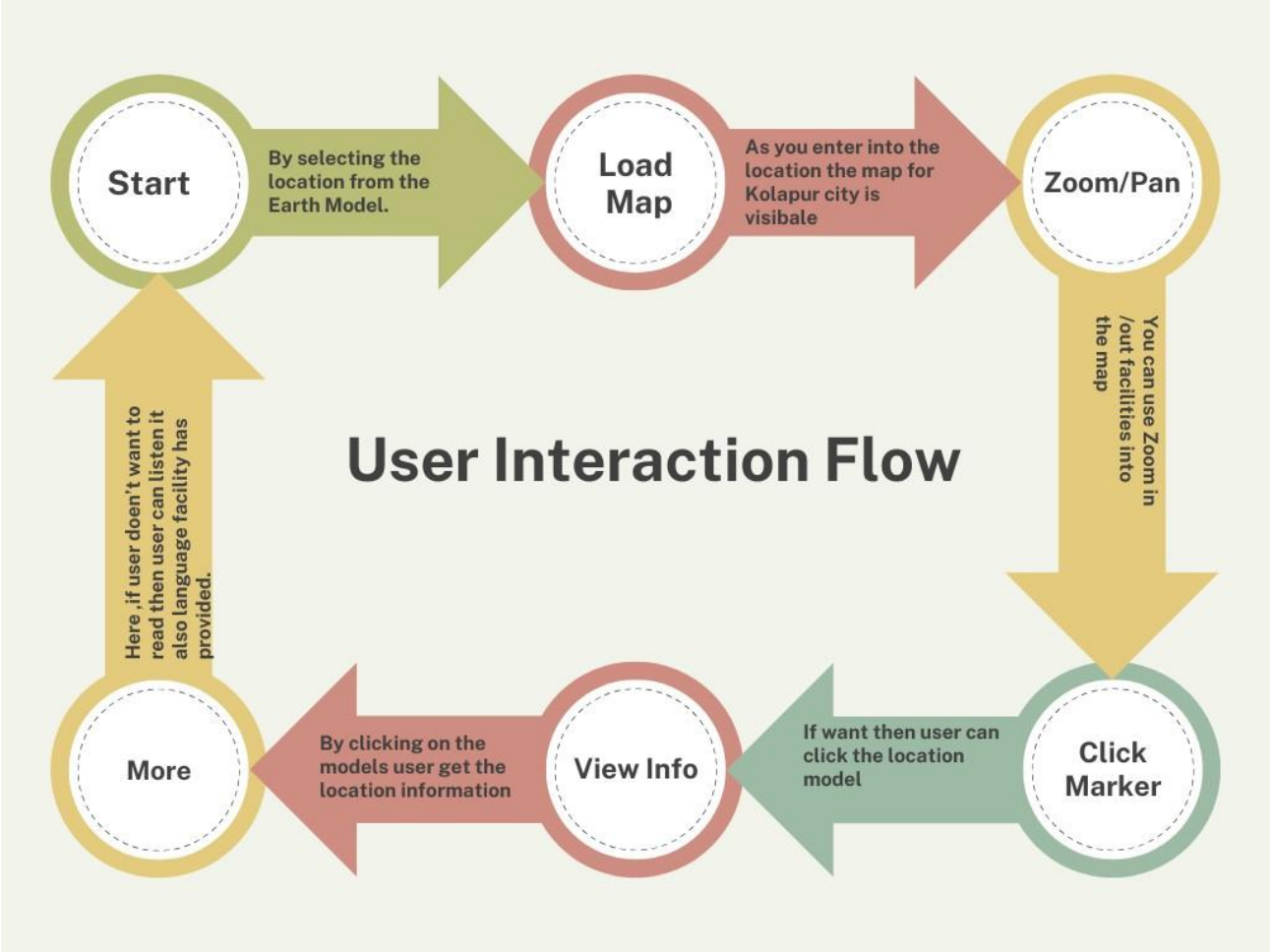
To visualize the system structure, **flowcharts**, **Data Flow Diagrams (DFD)**, and **UML diagrams** were created. The Level-0 DFD shows how user input (clicks, selections) passes through the system and interacts with APIs to produce the desired output. The flowchart explains the process of navigating from the home screen to viewing detailed landmark information.

The UML diagrams further describe class structures and module relationships. These tools helped in organizing the project's logic and ensuring data integrity throughout the workflow. The combination of these visual representations allowed developers to better plan data flow and optimize performance.

3.4 User Interface Design

The **User Interface (UI)** design emphasizes simplicity and usability. Built using **React**, it adapts fluidly to various devices. Each element—buttons, cards, and pop-ups—follows a consistent design language with soft colors and rounded edges to improve aesthetics. The interface offers toggles for **language selection** and **theme switching**, allowing users to personalize their experience.

Attention was also given to accessibility by incorporating readable fonts, adequate color contrast, and intuitive iconography. The placement of map controls, search bar, and navigation menu ensures minimal clutter. The end result is a user interface that feels professional, modern, and responsive to user needs.



3.5 Database Design

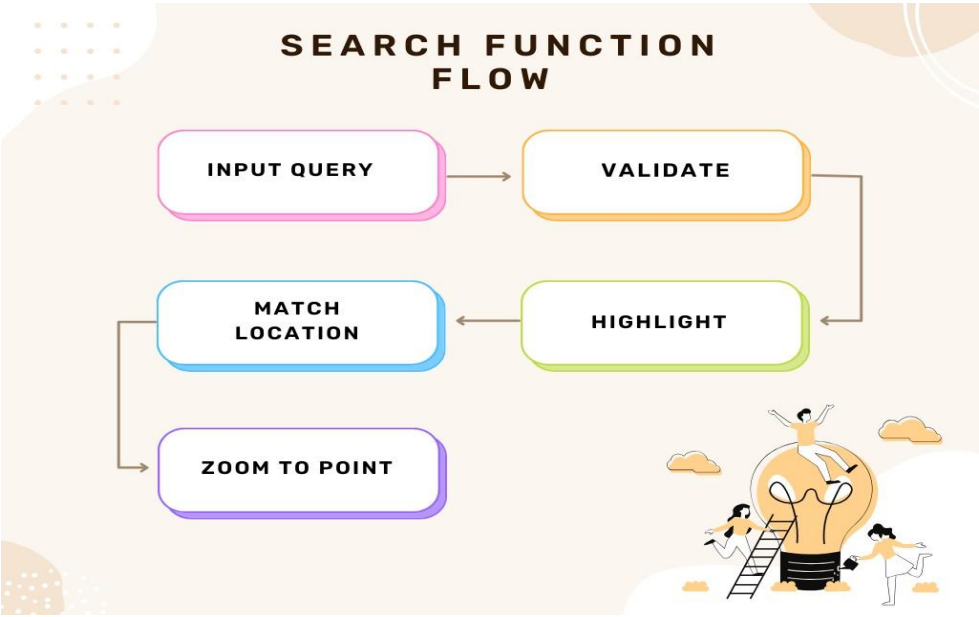
The system uses a **JSON-based or lightweight database** to store landmark details, translations, and audio links. The ER diagram consists of tables like Location, LanguageInfo, and AudioFiles. Each landmark entry includes attributes like ID, name, coordinates, description, and media references.

This database structure supports scalability—new landmarks or languages can easily be added without major code modifications. Data is fetched asynchronously to improve speed and reduce latency. Effective database design ensures seamless integration with the frontend while maintaining performance and consistency.

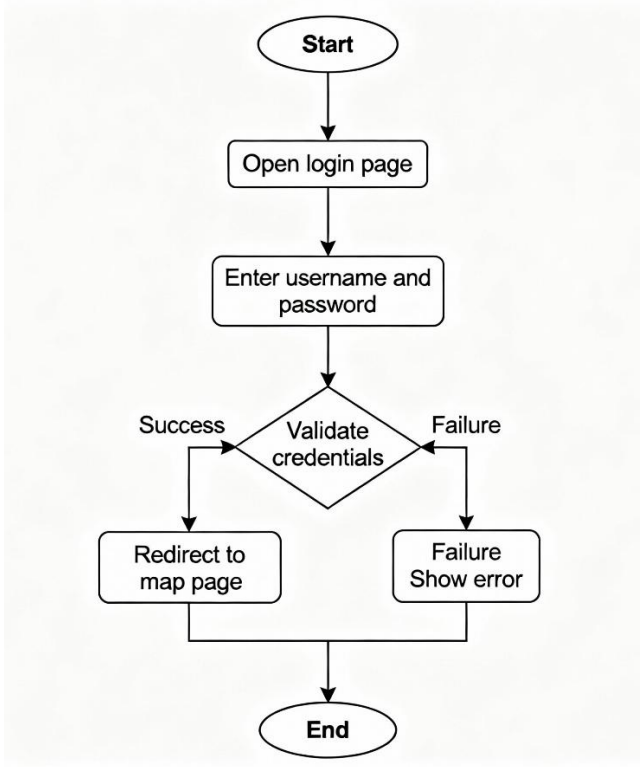
3.6 Methodology

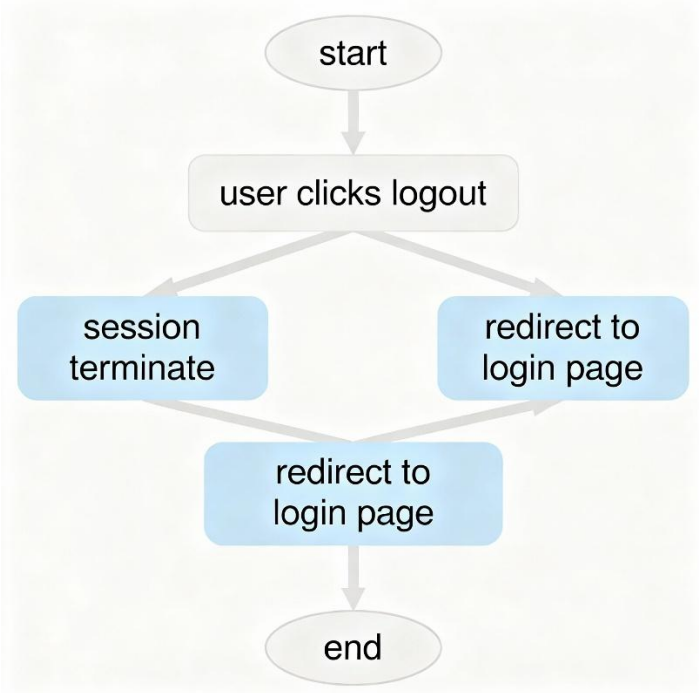
The methodology combines **front-end, 3D modeling, and mapping technologies**. Mapbox API provides the base map and handles geographic rendering, while Blender creates 3D models of major landmarks. These models are then integrated into the web interface through React components. The system also embeds Google Street View for real-world visualization.

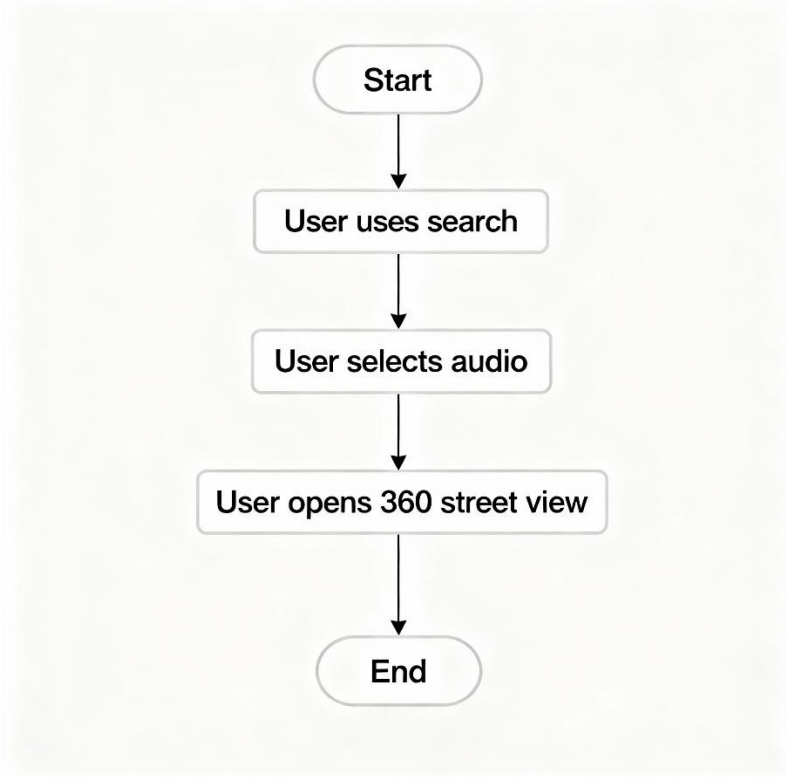
The workflow involves designing models in Blender, uploading them to the web environment, and synchronizing them with location data in Mapbox. This combination ensures a balance between realism and performance. The chosen methodology supports easy maintenance, modular updates, and efficient user experience delivery.

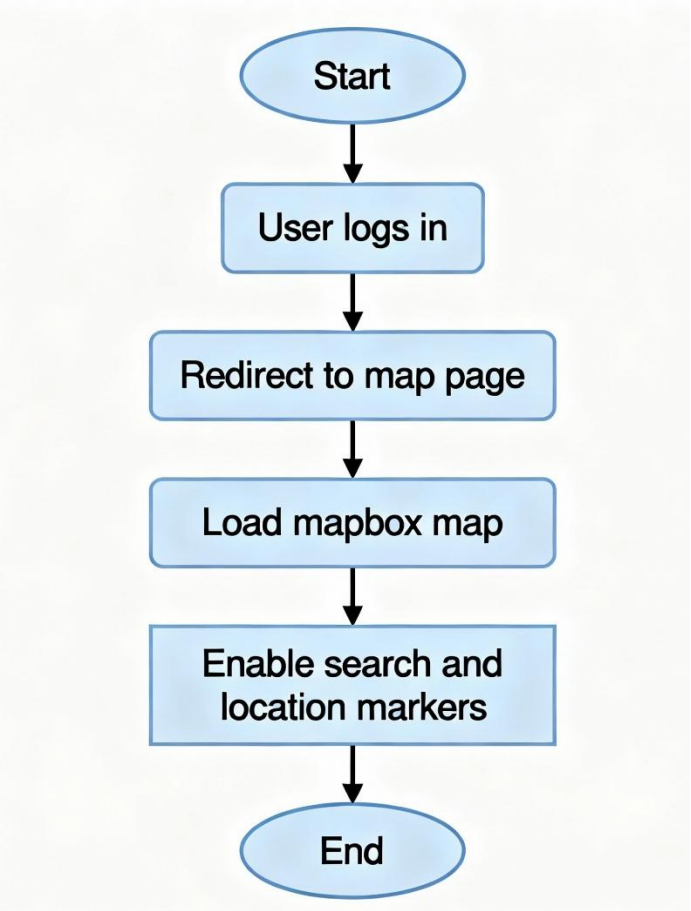


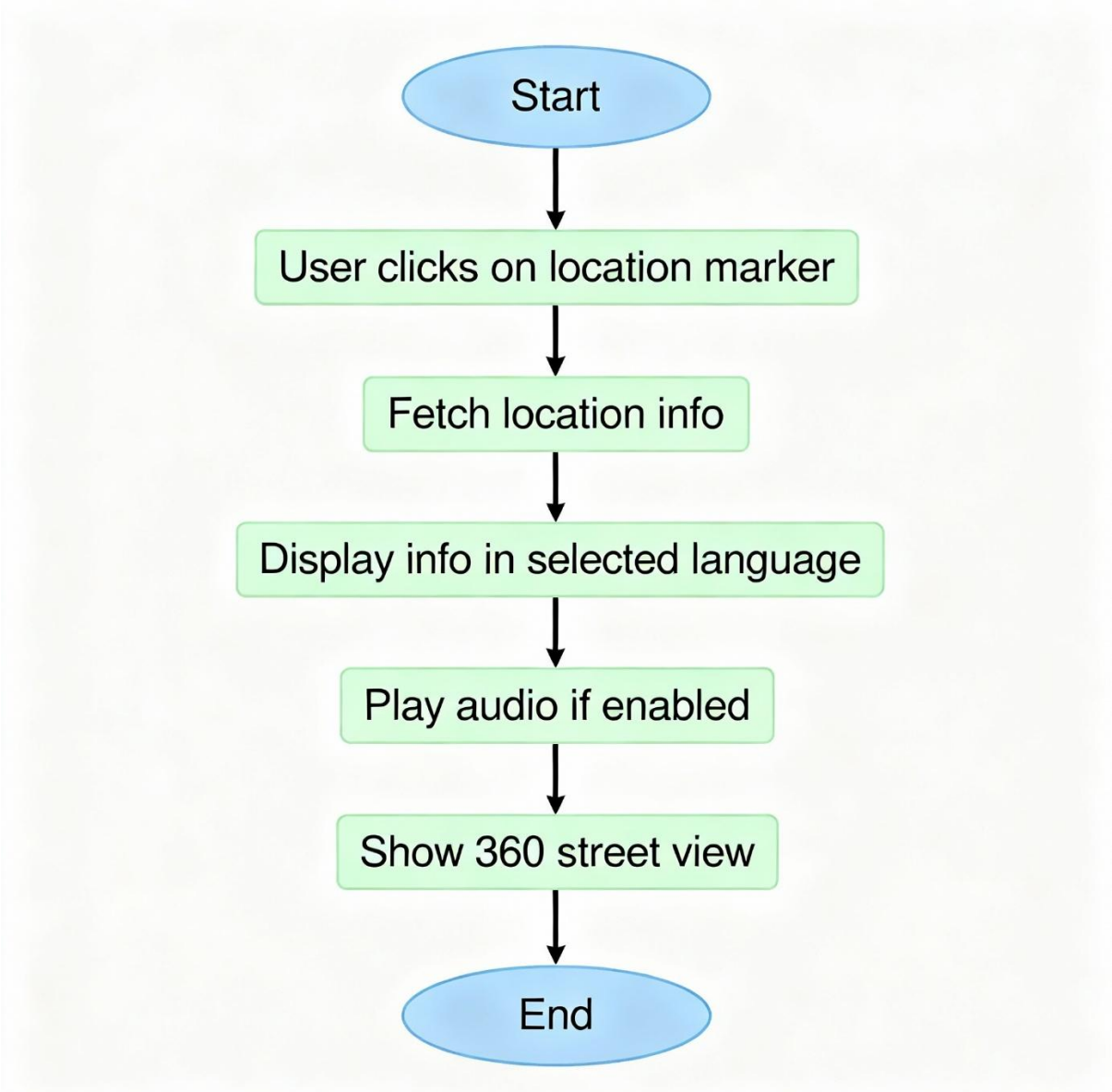
System Design Flowcharts






Start
User uses search
User selects audio
User opens 360 street view
End


Start
User logs in
Redirect to map page
Load mapbox map
Enable search and location markers
End


Start
User clicks on location marker
Fetch location info
Display info in selected language
Play audio if enabled
Show 360 street view
End

Chapter 4

IMPLEMENTATION, TESTING & MAINTENANCE

4.1 Implementation Tools

Implementation involves multiple tools working in coordination. **HTML, CSS, and JavaScript** form the foundation, while **React.js** manages dynamic components and state transitions. **Blender** is used for modeling landmarks, **Mapbox Studio** for customizing maps, and **Google Street View API** for embedding panoramic views. The development process is managed in **Visual Studio Code**, a lightweight yet powerful IDE.

These tools collectively ensure scalability and high performance. Each was selected for its specific capability—Blender for graphics, Mapbox for geospatial control, and React for efficient rendering. Together, they create a cohesive system capable of delivering smooth animations and interactive experiences.

4.2 Coding Standards

The project follows **consistent coding standards** to maintain readability and scalability. Each React component has meaningful names, proper indentation, and structured comments. CSS is modularized to maintain clean styling, and JavaScript code adheres to ES6 syntax for better performance and compatibility.

Reusable components are prioritized to avoid redundancy. This ensures that updates in one section propagate automatically across related modules. Following such standards reduces errors, facilitates debugging, and keeps the system maintainable for future developers.

4.3 Project Scheduling

Effective **time management** played a crucial role in successful implementation. Tools like **Trello** and **Gantt charts** were used for planning and tracking tasks. Each project phase—analysis, design, implementation, and testing—was allotted specific timelines to maintain progress consistency.

The use of Gantt charts provided a clear visual representation of task dependencies, milestones, and deadlines. This structured scheduling ensured smooth coordination between team members and timely completion of the project within the proposed timeline.

4.4 Testing

Testing was performed in several stages—**unit testing**, **integration testing**, and **usability testing**. Unit tests checked individual functionalities such as map rendering, audio playback, and language switching. Integration tests verified the synchronization between components, ensuring data and visuals worked seamlessly together.

Usability testing was conducted with real users to gather feedback on accessibility and navigation ease. Multiple browsers and devices were tested to confirm performance stability. The outcome demonstrated that the system runs efficiently, with smooth transitions and accurate responses, fulfilling all expected requirements.

Chapter 5

RESULTS AND DISCUSSIONS

5.1 User Interface Representation

The final user interface presents a visually appealing, responsive, and interactive environment. The **Home Page** greets users with a short animation and a start button leading to the 3D map interface. The **Map Page** displays the entire Kolhapur district, allowing users to zoom, rotate, and click on landmarks. Each click triggers an information panel with text and audio narration.

The UI emphasizes clarity and engagement, with easy navigation and multilingual accessibility. Smooth transitions, thematic backgrounds, and high-quality images make the interface enjoyable to explore. Overall, the UI meets both functional and aesthetic expectations.

5.1.1 Module Description

The system includes four main modules:

1. **Map Interface** – Handles 3D visualization of Kolhapur district using Mapbox.
2. **Location Info Display** – Provides landmark-specific information with text and images.
3. **Audio & Language System** – Offers multilingual audio playback to enhance accessibility.
4. **Street View Integration** – Embeds Google Street View for realistic, on-site visualization.

Each module works collaboratively to deliver an immersive experience. The modular design also enables easy modification and expansion, allowing the system to scale beyond Kolhapur in the future.

5.2 Snapshots of System

Snapshots include the **Home Page**, **Map Interface**, **Information Page**, and **Street View Page**. These screenshots serve as visual proof of functionality and aesthetic

quality. They also demonstrate the transition between modules and the integration of 3D visuals with real-world imagery.

Each snapshot highlights specific functionality—for example, clicking a landmark and hearing an audio description or switching the theme mode. These images collectively validate the system's performance and usability.

5.3 Back-End Representation

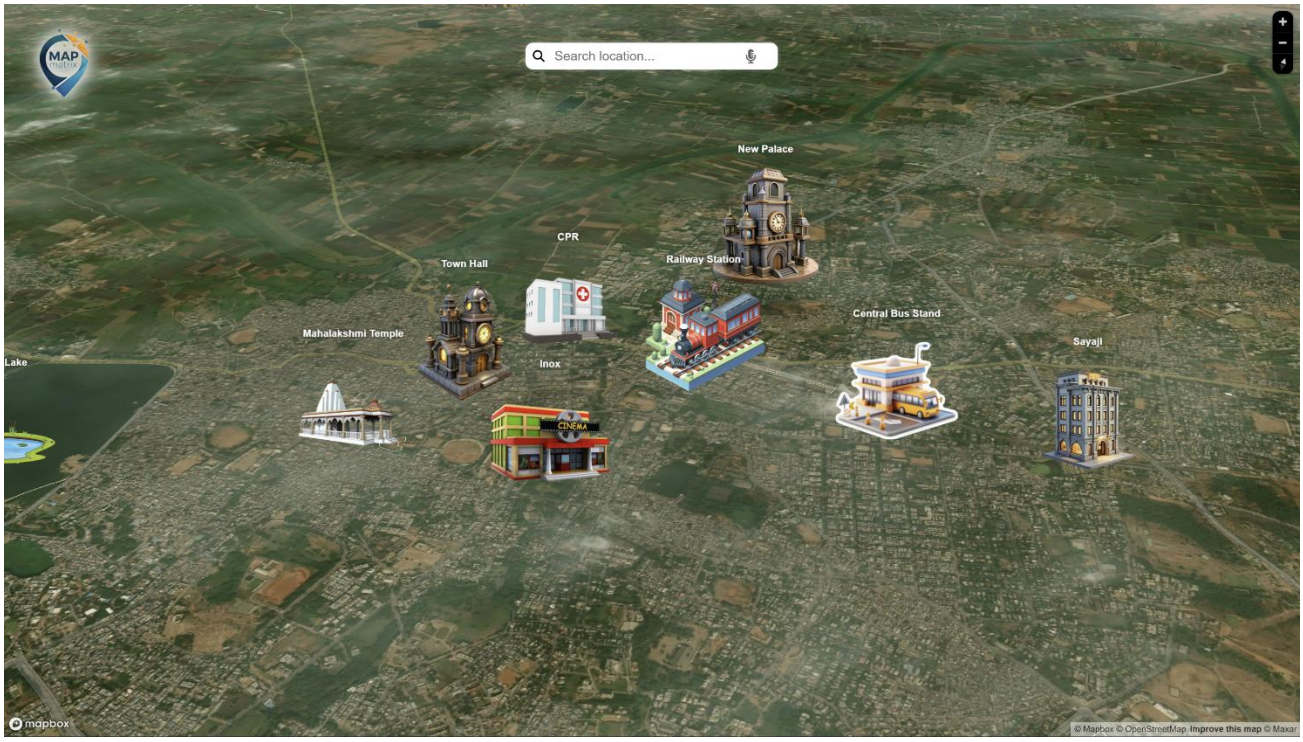
The back-end system uses lightweight JSON or a simple database to store and retrieve landmark data efficiently. Each record holds multilingual text and corresponding audio links. The data structure supports quick access and flexible modification.

The back-end architecture ensures stability and reliability while keeping the system light and browser-friendly. The **ER diagram** and **data flow representations** depict how user requests are processed, ensuring transparency in data handling and logic.

Home Page

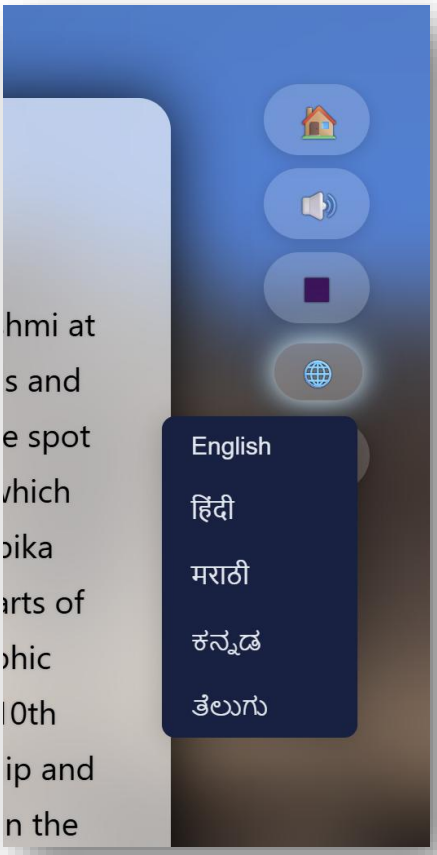


Map Page



Information Page





Multilingual & audio support

Search and Filter Option



Chapter 6

CONCLUSION AND FUTURE SCOPE

Conclusion

The project successfully develops an **Interactive 3D Map of Kolhapur District** that combines web mapping, 3D modeling, and multilingual accessibility into a single cohesive platform. It provides users with a dynamic, informative, and immersive digital experience that promotes both tourism and cultural education.

By integrating 3D visualization, multiple languages, and audio narration, the system achieves inclusivity and modernity. It transforms traditional map usage into an engaging digital journey, highlighting the power of web-based visualization in local and regional development.

Future Scope

Future enhancements can include **navigation pathfinding**, **real-time weather updates**, and **live camera feeds** from major landmarks. Expanding the project to include more districts or even the entire state can make it a large-scale tourism and educational resource.

A **mobile application version** could also be developed to increase accessibility and usability for on-the-go users. Integration of augmented reality (AR) could further enrich the exploration experience, allowing users to interact with landmarks virtually. With such future improvements, this project holds immense potential for scalability and innovation.

Mobile Application:

In the future, the interactive 3D map can be extended into a **mobile application** to enhance accessibility and usability. A mobile version will allow users to explore the Kolhapur district conveniently from their smartphones or tablets, without relying on desktop browsers. It can include GPS-based tracking to help users identify their real-time location within the district and navigate easily to nearby landmarks. The mobile

app can also support offline functionality, enabling users to access basic map data even without an active internet connection, which is particularly beneficial for travelers in remote areas.

AI Route Suggestions:

Integrating **Artificial Intelligence (AI)** into the system can greatly improve navigation and personalization. AI algorithms can analyze user behavior, travel patterns, and preferences to suggest optimized routes between multiple destinations. For example, tourists can receive customized itineraries based on their interests—such as temples, museums, or natural spots. Additionally, AI can suggest less crowded paths or alternative routes during peak hours, thereby improving travel efficiency and enhancing the overall user experience.

Real-Time Updates:

Implementing **real-time updates** would make the system more dynamic and useful for everyday users. This feature would enable the map to display live information such as traffic conditions, road closures, local events, and weather updates. Real-time data integration can be achieved using APIs and cloud-based services that continuously refresh information. This would transform the 3D map from a static visualization into an interactive, constantly evolving platform that provides accurate and timely insights for users exploring the Kolhapur district.

Cloud Analytics:

The integration of **cloud analytics** can provide valuable insights into user engagement and system performance. Data collected from user interactions—such as frequently visited landmarks, popular routes, or preferred languages—can be stored and analyzed on cloud platforms. This analysis would help developers and administrators optimize the platform, improve content, and make data-driven decisions for future updates. Moreover, cloud storage would ensure scalability, allowing the system to handle large amounts of multimedia and user data efficiently without compromising speed or reliability.

Holographic Maps:

A futuristic extension of the project could include **holographic or augmented reality (AR) maps**. By combining 3D projection and AR technologies, users could interact with lifelike holographic models of Kolhapur's landmarks. This would offer a next-

level immersive experience, especially for educational and tourism purposes. Users could explore structures from different angles, view scaled models in real space, and even experience virtual tours through holographic interfaces. Such innovation would set a new standard for digital mapping and virtual tourism.

Holographic map with AI support



REFERENCES / BIBLIOGRAPHY

In order to successfully complete the project, a variety of **online and technical resources** were referred to throughout the design and implementation process. These references provided valuable insights into technologies such as React.js, Mapbox, Blender, and Google Maps APIs. Documentation and tutorials from these official sources guided the integration of 3D models, map layers, and interactive features. Additionally, open data repositories and academic articles on digital mapping and visualization techniques helped refine the conceptual and design approach.
The research materials consulted were not limited to technical documents alone but also included **online forums, community discussions, and developer blogs**. These platforms offered practical solutions to issues encountered during development, such as optimizing 3D performance, managing API keys, and enabling multilingual support. The combination of formal documentation and community knowledge ensured both theoretical understanding and hands-on problem-solving.
Moreover, various **open data sources and government websites** were used to collect location-specific information about Kolhapur's geography, landmarks, and cultural sites. Accurate and verified data were essential to ensure reliability in visualization and factual correctness in the displayed content. The use of diverse and authentic sources ensured the project maintained a balance between technical excellence and informational accuracy.

List of References:

1. **Mapbox Documentation** – https://www.mapbox.com
   Provided detailed guidance on integrating 3D maps, custom map layers, and API usage for rendering Kolhapur district visualization.

2. **Blender Official Website** – https://www.blender.org
   Offered tutorials and resources for creating, exporting, and optimizing 3D landmark models for web integration.

3. **React.js Official Documentation** – https://react.dev
   Served as the primary reference for building the front-end interface, managing component state, and ensuring responsiveness.

4. **Google Maps API Documentation** – https://developers.google.com/maps
   Helped in embedding 360° Street Views and understanding geolocation APIs used in the project.

5. **W3Schools and MDN Web Docs** – https://www.w3schools.com, https://developer.mozilla.org
   Provided supplementary information on HTML, CSS, and JavaScript syntax and functionalities used in front-end design.

6. **Open Data Sources on Kolhapur** – Government of Maharashtra tourism and geography portals, Wikipedia, and local heritage archives.
   Supplied factual and cultural information about landmarks to ensure accurate representation within the 3D map.
